Arisha Firoz

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Career Objective

Motivated Computer Science undergraduate with hands-on experience in full-stack web development, REST API integration, and AI-driven applications. Backed by strong core computer science fundamentals and 235+ solved LeetCode problems. Eager to mark my contribution by engineering scalable, high-performance software solutions.

Education

United College of Engineering & Research, Prayagraj 2023 – Present
B.Tech in Computer Science CGPA: 7.7
St. Anthony's Convent Girls Inter College 2020 – 2022
High School & Intermediate (Science) 84.17% | 69.6%

Experience

Frontend Developer Intern | Construction Builder Solution Mar 2026 – May 2026

- Engineered responsive single-page interfaces using **AngularJS**, improving overall web UI responsiveness and **user experience**.
- Integrated frontend components with backend services using **REST APIs** and **JSON**, reducing client-server payload communication latency.
- Conducted **cross-browser compatibility testing, bug fixing,** and performance optimization across core platform pages.

Frontend Development Intern | E-Commerce Platform Jan 2026 – Mar 2026

- Designed and deployed modular, reusable UI components using **HTML5**, **CSS3**, **JavaScript** and **Bootstrap**.
- Collaborated with backend developers to integrate **third-party APIs, conduct UI testing,** and resolve cross-device rendering issues.

Projects

AI-Powered Code Review Agent | FastAPI, Python, Groq LLM, GitHub REST API | Live Link

- Built an **automated AI code reviewer** that analyzes **GitHub Pull Request diffs** and posts targeted **code quality feedback**.
- Architected a **multi-agent orchestration pipeline** consisting of **Static Analysis**, **Security Audit**, **Code Style**, and **Executive Summary agents**.
- Optimized the **prompt context window** by feeding **diff-only payloads**, minimizing **token usage** and **API overhead**.

Real-Time YouTube Watch Party | React, Node.js, TypeScript, Express, Socket.IO

- Built a **real-time watch-party app** where a Host or Moderator's play, pause, seek and video-change actions stay **synchronized across all participants over WebSockets**, with late joiners starting at the live playback position.
- Designed a **server-authoritative architecture** (`Room` and `RoomManager` classes) with **backend-enforced role-based access control** (Host, Moderator, Participant) and automatic host succession.
- Validated all payloads and rate-limited chat and reactions; wrote **attack scripts** confirming that raw socket events from unauthorized users are rejected. Deployed on **Vercel** (frontend) and **Render** (backend).

Predictive Maintenance – RUL Prediction | Python, Scikit-learn, Random Forest, Pandas | Live Link

- Engineered **60+ features** from 26 raw NASA C-MAPSS turbofan engine sensors using **rolling mean, rolling std,** and **rate-of-change signals** to capture degradation patterns, reducing Random Forest RMSE from 40.76 to **35.32**.
- Built a **model progression pipeline** — Linear Regression (baseline RMSE: 43.87) to Random Forest — with **correlation-based feature selection** and **StandardScaler preprocessing** to predict engine Remaining Useful Life (RUL) before failure.
- Self-engineered the **RUL target variable** from raw sensor data; cost simulation across **518 critical engines** projected **Rs. 21.7 Cr in prevented breakdown costs**, demonstrating end-to-end **ML problem formulation**.

E-Commerce Web Application | Django, JavaScript | Live Link

- Implemented **secure user authentication** using JWT/Sessions and integrated **payment gateway flows** for seamless checkout processing.
- Developed **automated inventory tracking** and asynchronous **AJAX-based dynamic shopping cart workflows** for efficient order management.

Achievements

- **Competitive Programming:** Solved **235+ Data Structures & Algorithms problems** on LeetCode.
- **Flipkart GRID 7.0:** Successfully cleared **Round 1** in the **Software Development Track**; qualified for the **Amazon ML Summer School**.
- **Smart India Hackathon (SIH) 2025:** Secured position in **Top 13** out of 90+ teams.
- **Social Summer of Code (SSOC) 2025:** Secured **Global Rank 47** among **350+ open-source contributors**.
- **HackQuest 2025:** Selected to showcase the **AlertNexa project** to an industry panel.
- **Certifications:** Java (Bronze Level) – HackerRank.

Technical Skills

Languages: Python, Java, C, JavaScript
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Web Development: HTML5, CSS3, Bootstrap, AngularJS, React, FastAPI, Django, REST APIs
Databases: SQL, MongoDB
Tools: Git, GitHub, Docker
AI/ML: Scikit-learn, Random Forest, Feature Engineering, RAG, Prompt Engineering